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1 "/Users/son/Library/Mobile Documents/com~apple~CloudDocs/University/Laboratory/Year 3 Lab/pycharm_venv/bin/python3" /Users/son/Library/Mobile Documents/com~apple~CloudDocs/University/Laboratory/Year 3 Lab/Data_Analysis/main.py
2 Enter the path to the video (with A0): /Users/son/Library/Mobile Documents/com~apple~CloudDocs/University/Laboratory/Year 3 Lab/Data_Analysis/Videos/without_A0/3ohm/0.9A_3R_without_A0.avi
3 Processing Video
4 Title: 0.9A_3R_without_A0, Current: 0.9, Resistance: 3, With A0: False
5 Enter the stage of the process: 1
6 Initial 300 frames extracted
7 Amplitude: 7152.008279973361, xo: 764.7756160169421, yo: 614.0555964136767, sigma_x: 2.6382628319425288, sigma_y: 7.178589884814176, theta: 0.21189014592743696, offset: 0.5249758749748009
8 Function:  $7152.008279973361 * \exp(-((\cos(\theta)^2)/(2*\sigma_x^2) + (\sin(\theta)^2)/(2*\sigma_y^2))*(x-xo)^2 + 2*(-\sin(2*\theta)/(4*\sigma_x^2) + \sin(2*\theta)/(4*\sigma_y^2))*(x-xo)*(y-yo) + ((\sin(\theta)^2)/(2*\sigma_x^2) + (\cos(\theta)^2)/(2*\sigma_y^2))*(y-yo)^2) + \text{offset}$ 
9 ROI defined
10 2025-01-21 10:16:08.209 python3[69224:53988491] +[IMKClient subclass]: chose IMKClient_Modern
11 2025-01-21 10:16:08.209 python3[69224:53988491] +[IMKInputSession subclass]: chose IMKInputSession_Modern
12 Processed 100 frames
13 SNR: 87.31292724609375
14 Processed 200 frames
15 SNR: 93.99615478515625
16 Processed 300 frames
17 SNR: 101.36669158935547
18 Processed 400 frames
19 SNR: 72.75399780273438
20 Processed 500 frames
21 SNR: 70.11465454101562
22 Processed 600 frames
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23 SNR: 79.81420135498047
24 Processed 700 frames
25 SNR: 55.47468185424805
26 Processed 800 frames
27 SNR: 91.18795013427734
28 Processed 900 frames
29 SNR: 64.43373107910156
30 Processed 1000 frames
31 SNR: 145.77850341796875
32 Processed 1100 frames
33 SNR: 95.88768005371094
34 Processed 1200 frames
35 SNR: 83.47989654541016
36 Processed 1300 frames
37 SNR: 83.15280151367188
38 Processed 1400 frames
39 SNR: 71.77732849121094
40 Processed 1500 frames
41 SNR: 74.98123931884766
42 Processed 1600 frames
43 SNR: 64.4690170288086
44 Processed 1700 frames
45 SNR: 94.04866790771484
46 Processed 1800 frames
47 SNR: 101.1701431274414
48 Processed 1900 frames
49 SNR: 78.06660461425781
50 Processed 2000 frames
51 SNR: 67.21366119384766
52 Processed 2100 frames
53 SNR: 53.38188934326172
54 Processed 2200 frames
55 SNR: 87.35353088378906
56 Processed 2300 frames
57 SNR: 53.85987854003906
58 Processed 2400 frames
59 SNR: 51.638607025146484
60 Processed 2500 frames
61 SNR: 74.3849105834961
62 Processed 2600 frames
63 SNR: 69.18030548095703

64 Processed 2700 frames
65 SNR: 58.52189254760742
66 Processed 2800 frames
67 SNR: 51.38662338256836
68 Processed 2900 frames
69 SNR: 117.62480926513672
70 Processed 3000 frames
71 SNR: 69.56055450439453
72 Processed 3100 frames
73 SNR: 46.41629409790039
74 Processed 3200 frames
75 SNR: 29.658008575439453
76 Processed 3300 frames
77 SNR: 55.02708435058594
78 Processed 3400 frames
79 SNR: 61.618370056152344
80 Processed 3500 frames
81 SNR: 53.395503997802734
82 Processed 3600 frames
83 SNR: 40.78065490722656
84 Processed 3700 frames
85 SNR: 47.05713653564453
86 Processed 3800 frames
87 SNR: 95.51802062988281
88 Processed 3900 frames
89 SNR: 41.230106353759766
90 Processed 4000 frames
91 SNR: 32.8048095703125
92 Processed 4100 frames
93 SNR: 25.688867568969727
94 Processed 4200 frames
95 SNR: 36.886016845703125
96 Processed 4300 frames
97 SNR: 43.88710021972656
98 Processed 4400 frames
99 SNR: 42.61600112915039
100 Processed 4500 frames
101 SNR: 45.03514099121094
102 Processed 4600 frames
103 SNR: 41.95568084716797
104 Processed 4700 frames

105 SNR: 31.404584884643555
106 Processed 4800 frames
107 SNR: 47.874515533447266
108 Processed 4900 frames
109 SNR: 35.83378601074219
110 Processed 5000 frames
111 SNR: 47.99418258666992
112 Processed 5100 frames
113 SNR: 40.167388916015625
114 Processed 5200 frames
115 SNR: 43.179508209228516
116 Processed 5300 frames
117 SNR: 44.93317794799805
118 Processed 5400 frames
119 SNR: 30.102954864501953
120 Processed 5500 frames
121 SNR: 38.96345901489258
122 Processed 5600 frames
123 SNR: 32.94062805175781
124 Processed 5700 frames
125 SNR: 74.72721099853516
126 Processed 5800 frames
127 SNR: 38.580284118652344
128 Processed 5900 frames
129 SNR: 31.89458656311035
130 Processed 6000 frames
131 SNR: 50.15486145019531
132 Processed 6100 frames
133 SNR: 54.96736526489258
134 Processed 6200 frames
135 SNR: 45.724090576171875
136 Processed 6300 frames
137 SNR: 77.72918701171875
138 Processed 6400 frames
139 SNR: 56.93907165527344
140 Processed 6500 frames
141 SNR: 55.66878128051758
142 Processed 6600 frames
143 SNR: 40.93606185913086
144 Processed 6700 frames
145 SNR: 49.98186111450195

146 Processed 6800 frames
147 SNR: 50.76592254638672
148 Processed 6900 frames
149 SNR: 47.895912170410156
150 Processed 7000 frames
151 SNR: 44.44645690917969
152 Processed 7100 frames
153 SNR: 53.583099365234375
154 Processed 7200 frames
155 SNR: 34.30228042602539
156 Processed 7300 frames
157 SNR: 77.02604675292969
158 Processed 7400 frames
159 SNR: 65.21829986572266
160 Processed 7500 frames
161 SNR: 61.35331726074219
162 Processed 7600 frames
163 SNR: 45.57994842529297
164 Processed 7700 frames
165 SNR: 53.165504455566406
166 Processed 7800 frames
167 SNR: 44.28394317626953
168 Processed 7900 frames
169 SNR: 66.60806274414062
170 Processed 8000 frames
171 SNR: 43.62421798706055
172 Processed 8100 frames
173 SNR: 30.540828704833984
174 Processed 8200 frames
175 SNR: 31.794363021850586
176 Processed 8300 frames
177 SNR: 60.209205627441406
178 Processed 8400 frames
179 SNR: 55.39657211303711
180 Processed 8500 frames
181 SNR: 79.16206359863281
182 Processed 8600 frames
183 SNR: 53.3624153137207
184 Processed 8700 frames
185 SNR: 43.777862548828125
186 Processed 8800 frames

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187 SNR: 50.0178337097168
188 Processed 8900 frames
189 SNR: 46.53661346435547
190 Processed 9000 frames
191 SNR: 41.57467269897461
192 Processed 9100 frames
193 SNR: 31.78133773803711
194 Processed 9200 frames
195 SNR: 18.5238037109375
196 Processed 9300 frames
197 SNR: 18.336219787597656
198 Processed 9400 frames
199 SNR: 16.574554443359375
200 Processed 9500 frames
201 SNR: 17.362091064453125
202 Processed 9600 frames
203 SNR: 30.089139938354492
204 Processed 9700 frames
205 SNR: 59.00888442993164
206 Saving Signals and Signal_errs
207 Saving Signal_errs
208 Saving Time
209 Saving SNRs
210 Saving SNR_errs
211 Rolling AVG = 11
212
213 Process finished with exit code 0
214
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